

31	27	26	25	24	20	19	15	14	12	11		7	6		0	
funct7				rs2		rs1		funct3		rd			opcode		R-type	
imm[11:0]						rs1		funct3		rd			opcode		I-type	
imm[11:5]				rs2		rs1		funct3		imm[4:0]			opcode		S-type	
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]			opcode		B-type	
imm[31:12]										rd			opcode		U-type	
imm[20 10:1 11 19:12]										rd			opcode		J-type	

RV32I Base Instruction Set						
imm[31:12]				rd	0110111	LUI
imm[31:12]				rd	0010111	AUIPC
imm[20 10:1 11 19:12]				rd	1101111	JAL
imm[11:0]		rs1	000	rd	1100111	JALR
imm[12 10:5]	rs2	rs1	000	imm[4:1 11]	1100011	BEQ
imm[12 10:5]	rs2	rs1	001	imm[4:1 11]	1100011	BNE
imm[12 10:5]	rs2	rs1	100	imm[4:1 11]	1100011	BLT
imm[12 10:5]	rs2	rs1	101	imm[4:1 11]	1100011	BGE
imm[12 10:5]	rs2	rs1	110	imm[4:1 11]	1100011	BLTU
imm[12 10:5]	rs2	rs1	111	imm[4:1 11]	1100011	BGEU
imm[11:0]		rs1	000	rd	0000011	LB
imm[11:0]		rs1	001	rd	0000011	LH
imm[11:0]		rs1	010	rd	0000011	LW
imm[11:0]		rs1	100	rd	0000011	LBU
imm[11:0]		rs1	101	rd	0000011	LHU
imm[11:5]	rs2	rs1	000	imm[4:0]	0100011	SB
imm[11:5]	rs2	rs1	001	imm[4:0]	0100011	SH
imm[11:5]	rs2	rs1	010	imm[4:0]	0100011	SW
imm[11:0]		rs1	000	rd	0010011	ADDI
imm[11:0]		rs1	010	rd	0010011	SLTI
imm[11:0]		rs1	011	rd	0010011	SLTIU
imm[11:0]		rs1	100	rd	0010011	XORI
imm[11:0]		rs1	110	rd	0010011	ORI
imm[11:0]		rs1	111	rd	0010011	ANDI
0000000	shamt	rs1	001	rd	0010011	SLLI
0000000	shamt	rs1	101	rd	0010011	SRLI
0100000	shamt	rs1	101	rd	0010011	SRAI
0000000	rs2	rs1	000	rd	0110011	ADD
0100000	rs2	rs1	000	rd	0110011	SUB
0000000	rs2	rs1	001	rd	0110011	SLL

0000000		rs2	rs1	010	rd	0110011	SLT
0000000		rs2	rs1	011	rd	0110011	SLTU
0000000		rs2	rs1	100	rd	0110011	XOR
0000000		rs2	rs1	101	rd	0110011	SRL
0100000		rs2	rs1	101	rd	0110011	SRA
0000000		rs2	rs1	110	rd	0110011	OR
0000000		rs2	rs1	111	rd	0110011	AND
fm	pred	succ	rs1	000	rd	0001111	FENCE
1000	0011	0011	00000	000	00000	0001111	FENCE.TSO
0000	0001	0000	00000	000	00000	0001111	PAUSE
0000000000000			00000	000	00000	1110011	ECALL
0000000000001			00000	000	00000	1110011	EBREAK

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV64I Base Instruction Set (in addition to RV32I)

imm[11:0]			rs1	110	rd	0000011	LWU
imm[11:0]			rs1	011	rd	0000011	LD
imm[11:5]		rs2	rs1	011	imm[4:0]	0100011	SD
000000	shamt		rs1	001	rd	0010011	SLLI
000000	shamt		rs1	101	rd	0010011	SRLI
010000	shamt		rs1	101	rd	0010011	SRAI
imm[11:0]			rs1	000	rd	0011011	ADDIW
0000000		shamt	rs1	001	rd	0011011	SLLIW
0000000		shamt	rs1	101	rd	0011011	SRLIW
0100000		shamt	rs1	101	rd	0011011	SRAIW
0000000		rs2	rs1	000	rd	0111011	ADDW
0100000		rs2	rs1	000	rd	0111011	SUBW
0000000		rs2	rs1	001	rd	0111011	SLLW
0000000		rs2	rs1	101	rd	0111011	SRLW
0100000		rs2	rs1	101	rd	0111011	SRAW

RV32/RV64 Zifencei Standard Extension

imm[11:0]				rs1		001		rd		0001111		FENCE.I	
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RV32/RV64 Zicsr Standard Extension

csr				rs1		001		rd		1110011		CSRRW	
csr				rs1		010		rd		1110011		CSRRS	
csr				rs1		011		rd		1110011		CSRRC	
csr				uimm		101		rd		1110011		CSRRWI	
csr				uimm		110		rd		1110011		CSRRSI	
csr				uimm		111		rd		1110011		CSRRCI	

RV32M Standard Extension

0000001				rs2		rs1		000		rd		0110011		MUL
0000001				rs2		rs1		001		rd		0110011		MULH
0000001				rs2		rs1		010		rd		0110011		MULHSU
0000001				rs2		rs1		011		rd		0110011		MULHU
0000001				rs2		rs1		100		rd		0110011		DIV
0000001				rs2		rs1		101		rd		0110011		DIVU

0000001	rs2	rs1	110	rd	0110011	REM
0000001	rs2	rs1	111	rd	0110011	REMU

RV64M Standard Extension (in addition to RV32M)						
0000001	rs2	rs1	000	rd	0111011	MULW
0000001	rs2	rs1	100	rd	0111011	DIVW
0000001	rs2	rs1	101	rd	0111011	DIVUW
0000001	rs2	rs1	110	rd	0111011	REMW
0000001	rs2	rs1	111	rd	0111011	REMUW